Predictive Insights into Banking Customer Churn

DVA Capstone 2 Project Report

Field	Details
Sector	Banking and Financial Services
Team	Section B - Group 09
Dataset	Bank Customer Churn Dataset - Kaggle source , data/churn_prediction.csv
Submission Date	April 29, 2026

Team Members

S. No.	Name	Role
1	Supreet	Team Lead & Project Manager
2	Priyansh	Dashboard Developer
3	Meet Ahuja	Analysis Lead & Report Lead
4	Ritesh	ETL Lead & Data Engineer
5	Prateek	Strategy Lead & Presenter
6	Yashpal	Data Lead & Cleaning Lead

1. Executive Summary

Customer churn is a direct revenue and relationship risk for banks. The objective of this capstone project is to identify customer segments more likely to churn, understand the behavioral and financial patterns behind churn, and translate these patterns into retention actions.

The analysis covers 28,382 customers. The overall churn rate is 18.53%, with 5,260 churned customers and 23,122 retained customers. Python was used for extraction, cleaning, feature engineering, EDA, and statistical testing. Tableau was used to convert the final analysis tables into decision-oriented dashboards.

- Low current balance is the strongest practical churn warning signal. Low-balance customers show a
  65.31% churn rate.
- Self-employed customers have the highest occupation-level churn rate among major groups at 19.84%.
- Adult customers have the highest age-group churn rate at 19.98%, while middle-age customers contribute the highest churn count.
- High and very high activity segments show elevated churn rates, suggesting some customers may move funds before exiting.
- Statistical testing confirms significant differences for balance, gender, and occupation-related behavior.

The report recommends a balance-protection program, behavior-triggered retention alerts, occupation-specific offers, and segmented campaigns measured through a churn monitoring dashboard.

2. Sector Context and Problem Statement

2.1 Sector Context

In retail banking, customer relationships are built over time through savings balances, deposits, transactions, and product engagement. Losing customers creates both immediate and long-term business risk: banks lose balances, cross-sell opportunities, product adoption potential, and relationship value.

Churn analytics helps banks move from reactive customer service to proactive retention. Instead of waiting until customers leave, analysts can identify patterns that indicate risk and recommend targeted interventions.

2.2 Business Problem

The business problem is to identify which customers are most likely to churn and what factors influence their decision. The project focuses on financial behavior, demographics, tenure, activity level, and customer segments.

2.3 Objectives

- Measure the overall churn rate and customer retention base.
- Identify high-risk customer segments by age, gender, occupation, balance, activity, and network category.
- Validate important churn relationships using statistical tests.
- Prepare Tableau-ready exports and dashboards for decision makers.
- Recommend practical retention strategies that a bank can act on.

3. Data Description

The dataset used for this project is a bank customer churn dataset sourced from Kaggle and stored locally in the repository as data/churn_prediction.csv. The cleaned analytical file is stored as data/churn_clean.csv.

Attribute	Description
Raw dataset size	28,382 rows x 21 columns
Cleaned dataset size	28,382 rows x 26 columns
Target column	churn in raw data; Churn in cleaned data before standardization
Target classes	0 = retained, 1 = churned
Time reference	last_transaction and derived transaction year/month/day fields
Primary file for dashboards	data/tableau_exports/churn_final_tableau.csv

3.1 Key Columns

Column Group	Examples	Analytical Use
Customer profile	age, gender, dependents, occupation	Segment-level churn comparison
Relationship tenure	vintage	Customer lifetime and loyalty proxy
Financial balance	current_balance, previous balance, average balance fields	Balance behavior and churn risk
Transaction behavior	current_month_credit, current_month_debit, total_transactions	Activity and money movement patterns
Target	churn / Churn	Binary churn outcome used for analysis

3.2 Data Quality Issues

- Missing values were present in dependents, city, gender, occupation, and last_transaction.
- Some monetary fields contained negative values and extreme outliers that required treatment.
- The dataset is anonymized; city and branch values are codes, not descriptive business locations.
- The dataset does not include product ownership, complaint history, customer income, or revenue value.

4. Data Cleaning and ETL Methodology

The ETL workflow was implemented in Python notebooks. The extraction notebook validates the raw file, the cleaning notebook prepares the analytical dataset, and the final load-prep notebook creates Tableau-ready exports.

Stage	Notebook	Main Output
Extract	01_extraction.ipynb	Raw dataset loaded and profiled
Clean and transform	02_cleaning.ipynb	Cleaned dataset saved as data/churn_clean.csv
Explore	03_eda.ipynb	Charts and segment-level findings
Statistical validation	04_statistical_analysis.ipynb	T-test, chi-square, ANOVA, correlation results
Final load prep	05_final_load_prep.ipynb	KPI summary tables and Tableau exports

4.1 Cleaning Steps

- Removed duplicate rows after validation. No duplicate rows were found.
- Dropped low-value identifier/location fields for modeling and analysis focus: customer_id, branch_code, and city.
- Filled missing gender with mode, dependents with median, and occupation with Unknown.
- Converted last_transaction into datetime and created transaction_year, transaction_month, transaction_day, and no_transaction_flag.
- Replaced negative monetary values with zero and capped monetary outliers at the 1st and 99th percentiles.
- Created business-friendly derived fields: age_group, balance_segment, total_transactions, and activity_level.

4.2 Assumptions

- Negative balances were treated as invalid for this capstone analysis and reset to zero.
- Outlier capping was used to reduce distortion from extreme balance and transaction values.
- Missing occupation was kept as Unknown instead of deleting records, preserving customer count.
- Vintage is treated as tenure in days and bucketed into month-equivalent groups for dashboard use.

5. KPI and Metric Framework

The KPI framework connects raw customer data to business questions. Each KPI is designed to support a decision on customer retention, segment targeting, or dashboard monitoring.

KPI	Formula / Logic	Decision Value
Churn Rate	Churned customers / total customers x 100	Measures overall retention risk
Total Customers	Count of all customer records	Defines population size
Total Churned	Sum of churn flag	Shows churn volume
Total Retained	Total customers - churned customers	Shows retained customer base
Avg Current Balance	Average current_balance by churn status	Identifies financial risk pattern
Avg Vintage	Average vintage by churn status	Compares relationship tenure
Avg Total Transactions	Average current credit + debit by churn status	Captures activity behavior
Segment Churn Rate	Churn rate by age, gender, occupation, balance, activity, NW category	Supports targeted actions

5.1 KPI Snapshot

Metric	Value
Total Customers	28,382.00
Total Churned	5,260.00
Total Retained	23,122.00
Overall Churn Rate %	18.53
Avg Current Balance - Churned	4,202.91
Avg Current Balance - Retained	6,734.15
Avg Vintage - Churned	2,088.42
Avg Vintage - Retained	2,091.76
Avg Total Transactions - Churned	7,908.48
Avg Total Transactions - Retained	3,515.49

6. Exploratory Data Analysis

EDA was used to identify major churn patterns before statistical testing. The analysis compared churn across customer profile variables, balance segments, activity levels, and tenure groups.

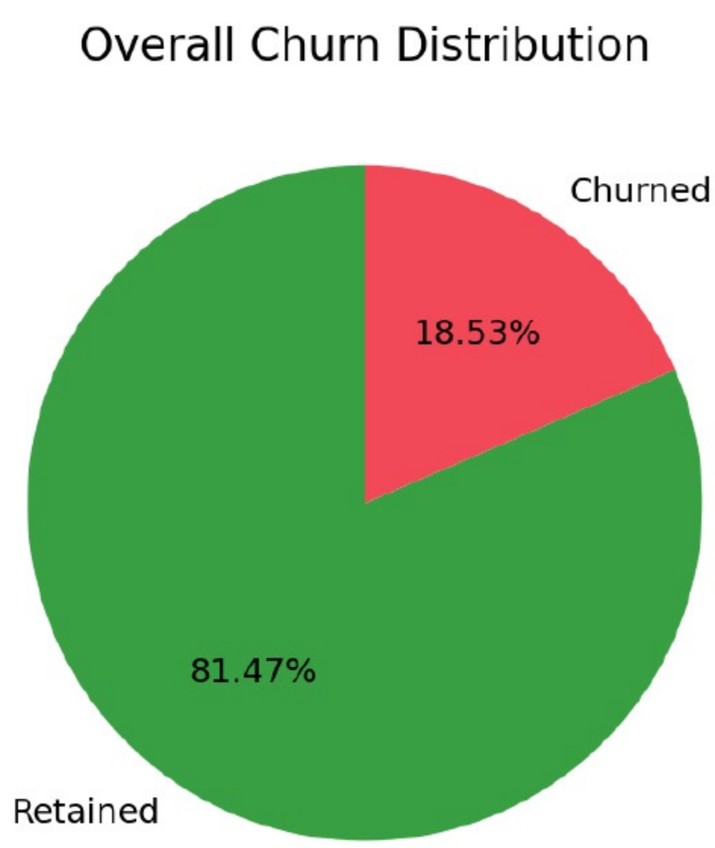
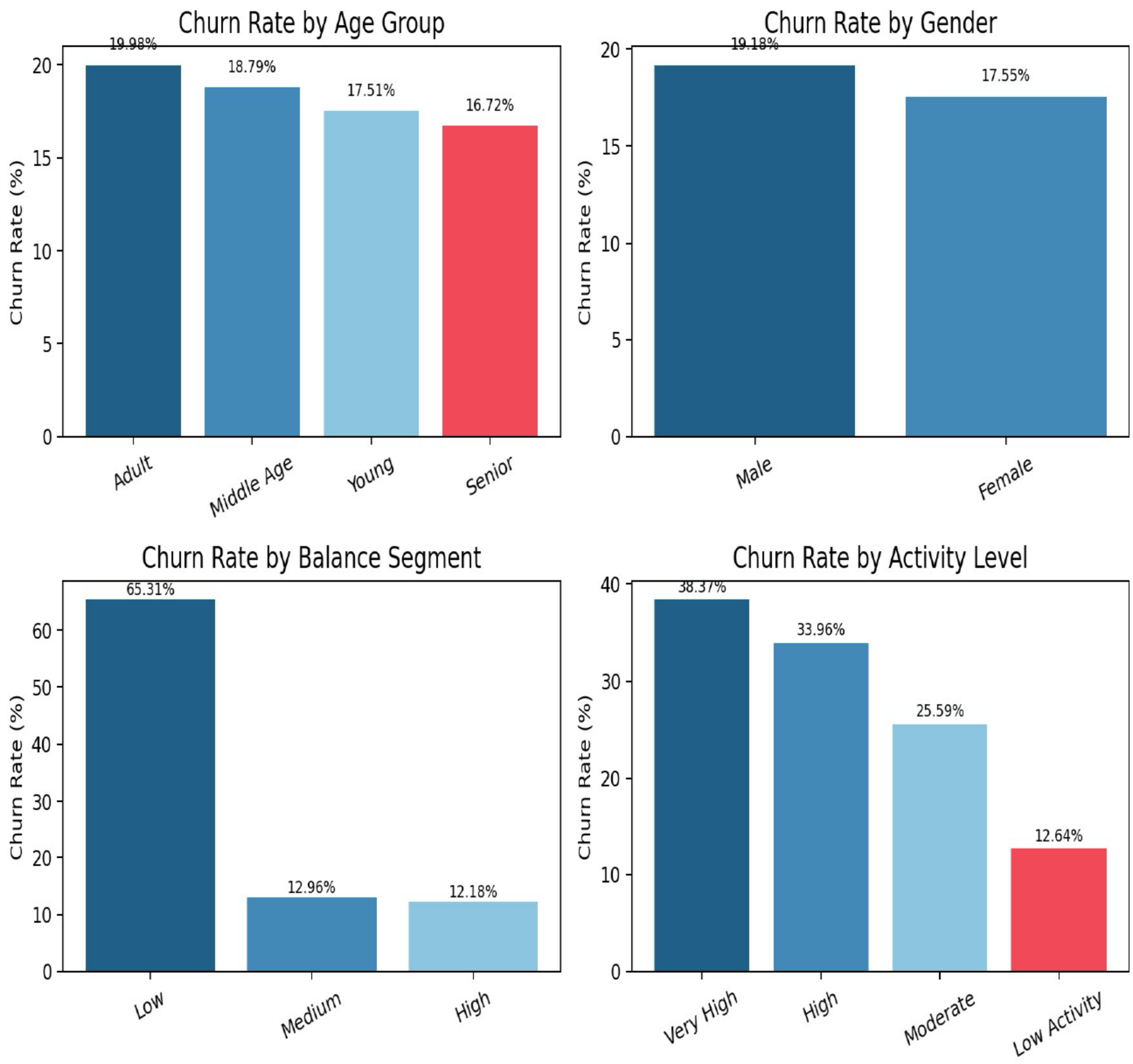


Figure 1: Overall churn distribution showing 18.53% churn and 81.47% retention.

The churn rate of 18.53% indicates that churn is a meaningful business issue. Nearly one in five customers in the dataset have churned, which is high enough to justify a structured retention program.

Figure 2: Churn rate comparison across age, ge



nder, balance, and activity segments.

The low-balance segment is the clearest high-risk group, with a churn rate of 65.31%. Adult customers have the highest age-group churn rate, while male customers churn slightly more than female customers.

6.1 Churn Volume and Business Priority

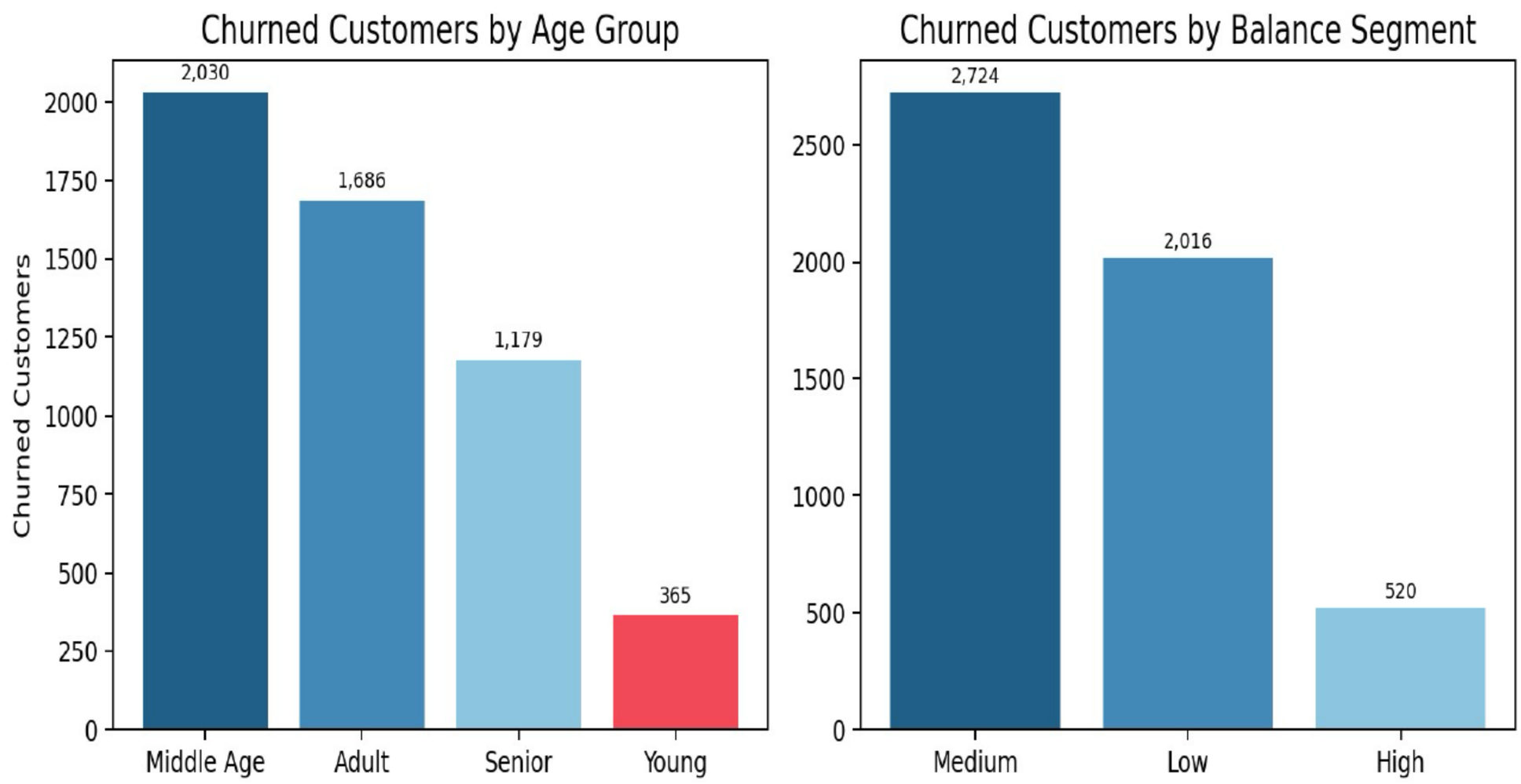


Figure 3: Churned customer counts by age group and balance segment.

Churn volume and churn rate should both be considered. Low-balance customers have the highest churn rate, but medium-balance customers contribute the largest churn count because they represent a much larger base.

Segment	Total Customers	Churned Customers	Churn Rate
High	4,271	520	12.18%
Low	3,087	2,016	65.31%
Medium	21,024	2,724	12.96%

7. Statistical Analysis Results

Statistical tests were used to validate whether observed differences in EDA are likely to be meaningful rather than random noise.

Test	Hypothesis / Question	Result	Business Interpretation
T-test: Balance vs Churn	Do churned and retained customers differ in current balance?	p = 5.61e-81, significant	Churned customers have significantly lower balances.
Chi-square: Gender vs Churn	Is churn independent of gender?	p = 0.00057, significant	Gender and churn are statistically related.
ANOVA: Occupation vs Balance	Does occupation affect balance maintained?	p = 1.20e-23, significant	Occupation groups differ significantly in maintained balance.
Correlation with Churn	Which numeric features move with churn?	current_month_debit = 0.186; current_balance = 0.105	Higher debit/activity and lower balance are useful risk signals.

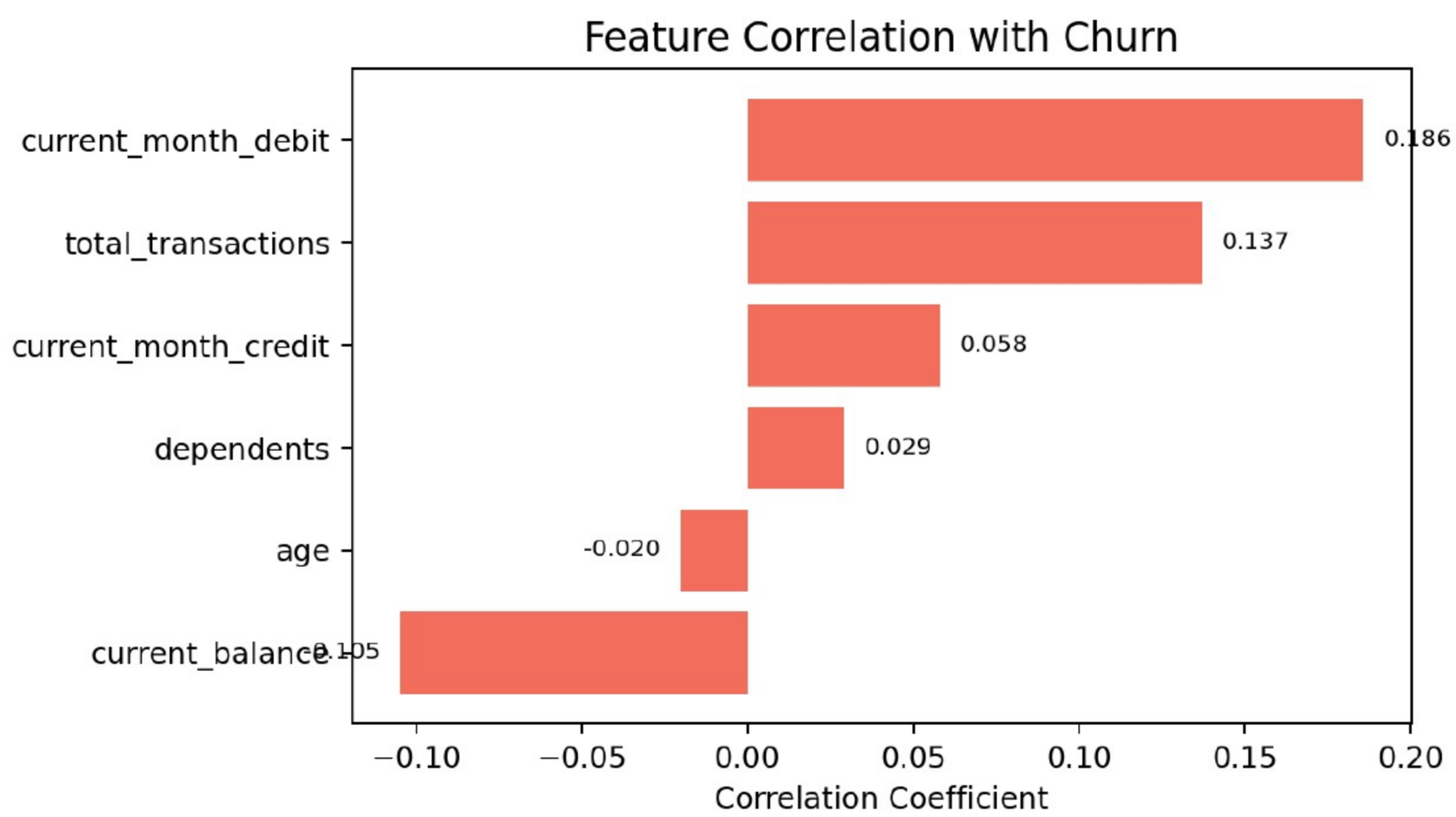


Figure 4: Feature correlation with churn.

The correlation analysis does not prove causation, but it helps prioritize practical monitoring variables. Current month debit and total transactions are positively related to churn, while current balance is negatively related to churn.

8. Dashboard Design and Walkthrough

The Tableau dashboard is designed for decision makers who need a quick executive overview and analysts who need to drill into risky customer segments. The dashboard uses KPI tiles, churn distribution views, segment filters, heatmaps, and bar charts.

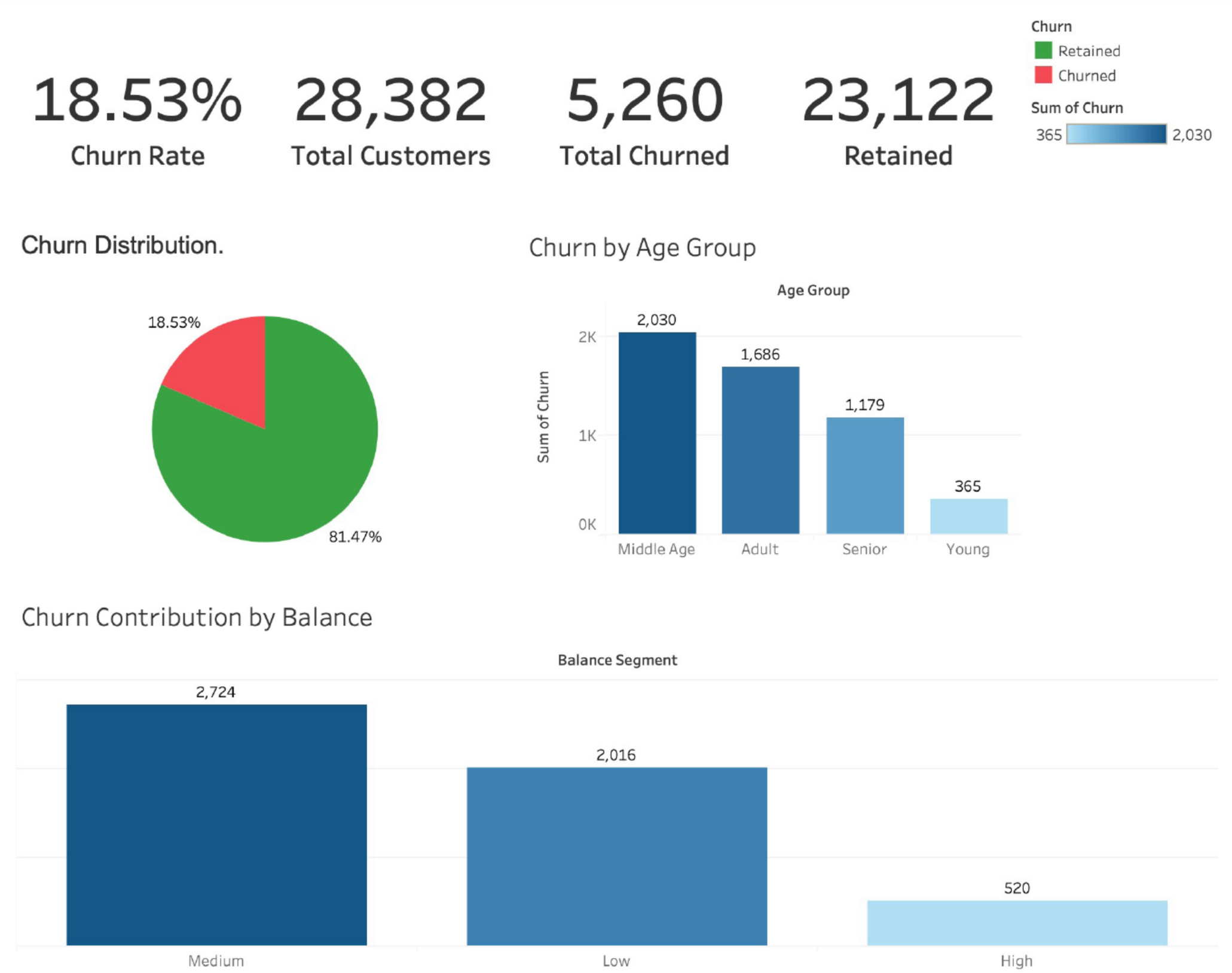


Figure 5: Executive dashboard view with overall churn KPIs and major churn distributions.

8.1 Executive View

- Shows the overall churn rate, total customers, churned customers, and retained customers.
- Highlights churn count by age group and balance segment.
- Helps leadership understand the size of the churn problem before moving to detailed segment analysis.

8.2 Segment Deep-Dive View

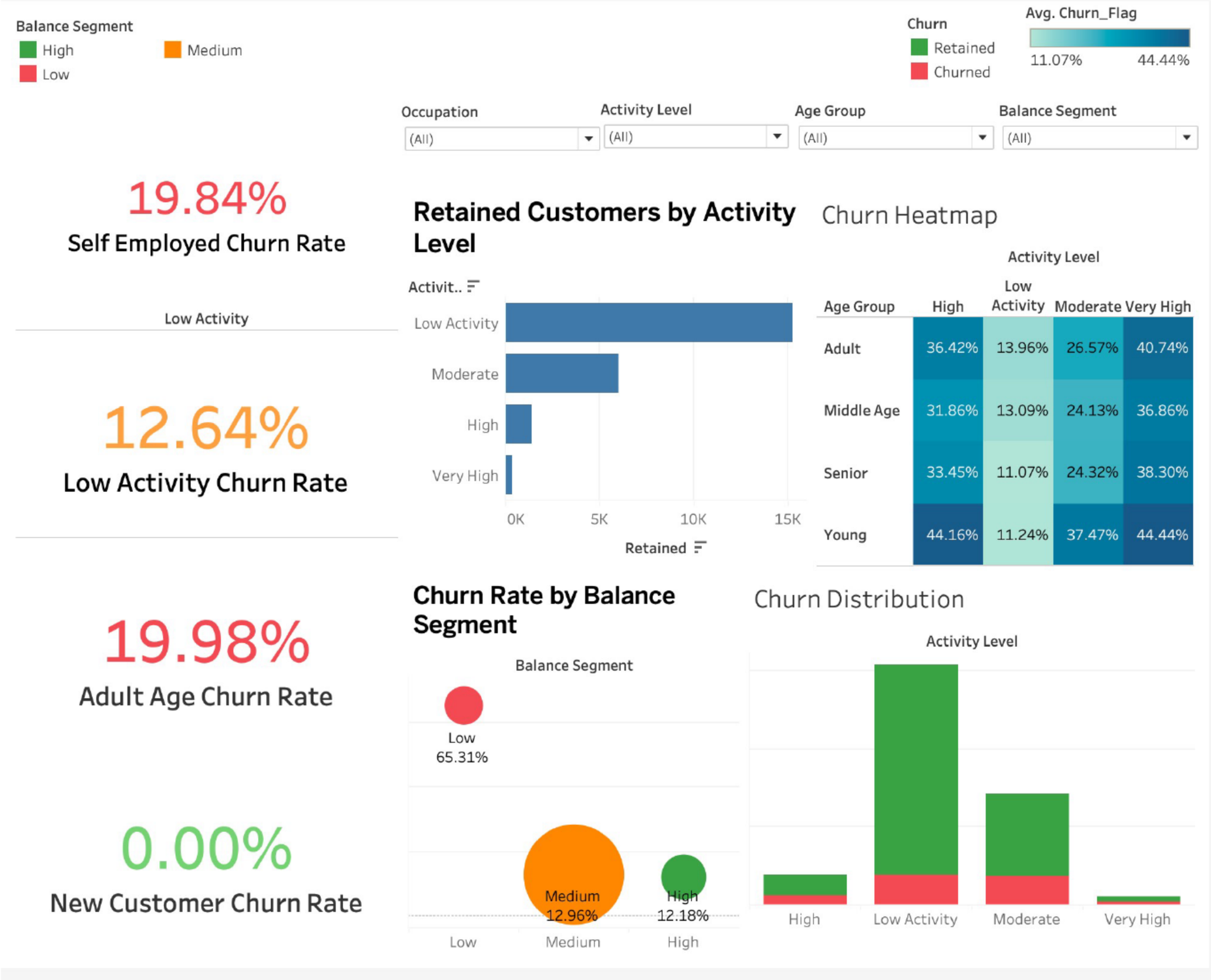


Figure 6: Segment deep-dive dashboard with filters, heatmap, activity view, and risk KPI cards.

- Filters allow slicing by occupation, age group, activity level, and balance segment.
- The heatmap compares age group and activity level to reveal concentrated churn-risk zones.
- KPI cards show priority segments such as self-employed churn rate, adult age churn rate, and low activity churn rate.

8.3 Operational Monitoring View

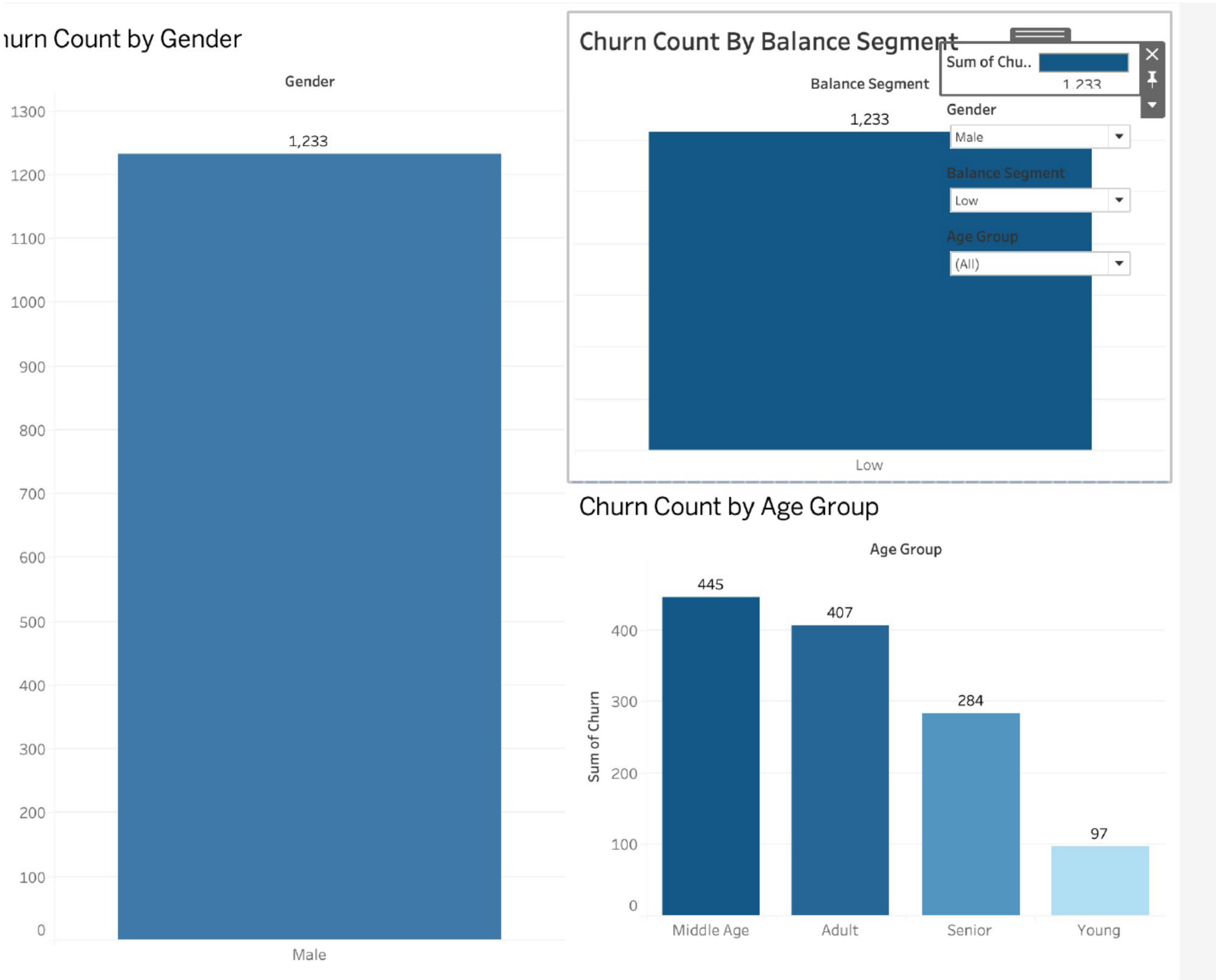


Figure 7: Operational dashboard view for monitoring gender, balance, and age-group churn counts.

This view supports operational analysis by showing churn distribution by gender, balance segment, and age group. It is useful for campaign planning because it displays both segment size and churn volume.

Key Insights in Decision Language

1. The overall churn rate is 18.53%, meaning nearly one out of five customers in the dataset has churned.
2. Low-balance customers are the most urgent risk segment, with a 65.31% churn rate.
3. Medium-balance customers contribute the largest churn volume: 2,724 churned customers.
4. Self-employed customers have the highest occupation-level churn rate among major occupation groups at 19.84%.
5. Adult customers have the highest age-group churn rate at 19.98%, while middle-age customers contribute the largest churn count.

6. Male customers show a slightly higher churn rate than female customers: 19.18% vs 17.55%.
7. Very high activity customers show a 38.37% churn rate, suggesting some high-activity customers may be moving money before exit.
8. Current month debit has the strongest positive correlation with churn among selected numeric variables.
9. Current balance has a negative relationship with churn, supporting low-balance monitoring as a retention trigger.
10. New customers show very low churn in the current vintage bucket, but the sample is too small to treat this as a stable conclusion.
11. Customer network category has similar churn rates across groups, so it is less actionable than balance, occupation, or activity behavior.
12. The dashboard should be used as an early-warning tool, not just a descriptive report.

9. Recommendations and Impact Estimation

Recommendation	Action	Estimated Impact
Launch a low-balance retention program	Target customers in the Low balance segment with balance-building nudges, fee waivers, micro-savings offers, and personal outreach.	A 10% relative reduction in low-balance churn could retain about 202 customers from the 2,016 churned low-balance group.
Create transaction-behavior alerts	Flag customers with sudden debit spikes or high total transactions, especially when balance is declining.	A 10% relative reduction among High and Very High activity churners could retain about 98 customers.
Design self-employed customer offers	Offer flexible current account benefits, overdraft education, business banking bundles, and advisory support.	A 5% relative reduction in self-employed churn could retain about 173 customers.
Segment campaigns by age and gender	Prioritize adult and male customer cohorts for communication testing while avoiding one-size-fitsall outreach.	Better targeting can reduce campaign waste and improve response rates.
Use Tableau dashboard as a monthly churn review tool	Refresh data and track churn KPIs by segment every month.	Creates a measurable retention workflow for managers and analysts.

Because the dataset does not contain customer revenue or product profitability, impact is estimated in retained-customer counts rather than monetary value. A future version can convert retained customers into financial impact once average revenue per customer is available.

10. Limitations and Future Scope

10.1 Limitations

- The dataset is anonymized and does not include branch names, customer income, product ownership, or complaint history.
- Churn is treated as a binary outcome, but the dataset does not explain the exact business definition of churn.
- The analysis is descriptive and statistical; it does not establish causal relationships.
- Some derived segments depend on chosen thresholds, such as balance and activity buckets.
- The dashboard reflects the available dataset and should be refreshed with recent operational data before production use.

12. Contribution Matrix

S. No.	Team Member Name	Role	GitHub Evidence / PR Link	Primary Contribution	Signature / Confirmation
1	**Supreet**	Team Lead & Project Manager	✅	Project Strategy, Leadership, Project Management & Oversight	Supreet
2	**Priyansh**	Dashboard Developer	✅	Dashboard Architecture, Tableau Development	Priyansh
3	**Meet Ahuja**	Analysis Lead & Report Lead	✅	Data Analysis, Report Writing	Meet Ahuja
4	**Ritesh**	ETL Lead & Data Engineer	✅	ETL Pipelines, Data Transformation	Ritesh
5	**Prateek**	Strategy Lead & Presenter	✅	Business Strategy Formulation, PPT Delivery	Prateek
6	**Yashpal**	Data Lead & Cleaning Lead	✅	Data Sourcing, Pre-processing, Data Quality	Yashpal